INTERNSHIP REPORT OF

Generative AI

Submitted To

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING IILM UNIVERSITY, GREATER NOIDA, U.P.

In partial fulfilment of the requirement of degree of

B.TECH (CSE)

PROJECTS UNDERTAKEN

1. Text Generation with GPT-2
2. Image Generation with Pre-trained Models
3. Text Generation with Markov Chains
4. Image-to-Image Translation with cGAN
5. Neural Style Transfer

By

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Batch: 2026-27

CANDIDATE'S DECLARATION

I, Abhay Srivastav, do hereby declare that the internship report titled “Generative AI Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

The work presented in this report is based on the practical learning and task specifications provided for the Generative AI internship. The report documents five assigned tasks covering text generation, image generation, Markov chains, image-to-image translation, and neural style transfer.

All references used in preparing this report have been acknowledged. I affirm that this report is my own work and has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement.

I further declare that I shall be solely responsible for any kind of copyright violation arising from the contents of this report.

Signature:

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INTERNSHIP COMPLETION CERTIFICATE

This page is reserved for the official internship completion certificate.

The original certificate may be inserted here without changing the report layout.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this internship report and the assigned Generative AI tasks.

I am thankful to Prodigy InfoTech for providing me with an opportunity to gain practical exposure to modern generative artificial intelligence concepts. The task set introduced applications of language models, image-generation systems, probabilistic text generation, conditional GANs, and neural style transfer.

I also express my sincere gratitude to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing the academic environment and support required to complete this work.

Finally, I am grateful to my faculty members, mentors, friends, and family for their valuable suggestions, encouragement, and support during the preparation of this report.

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4. PROJECT DESCRIPTION

4.1 Introduction

Generative Artificial Intelligence focuses on models that learn patterns from existing data and use those patterns to create new content. Depending on the model, the generated content may be text, images, or transformed visual representations.

The assigned internship work contains five practical tasks. The first explores text generation with GPT-2. The second explores image generation using pre-trained generative models. The third implements a statistical text generator using Markov chains. The fourth studies image-to-image translation with a conditional generative adversarial network called pix2pix. The fifth applies neural style transfer to combine the style of one image with the content of another.

Together, these tasks provide a compact practical study of several major ideas used in Generative AI: autoregressive language modelling, diffusion or pre-trained image generation, statistical sequence generation, adversarial image translation, and feature-based style transfer.

4.2 Objectives and Scope

- Understand the fundamentals of generative models and content generation.
- Explore text generation with GPT-2 using a custom dataset and fine-tuning workflow.
- Use a pre-trained image generation model to create images from text prompts.
- Implement a simple Markov-chain based text generator.
- Understand paired image-to-image translation using the pix2pix cGAN approach.
- Apply neural style transfer to combine content structure and artistic style.
- Document the assigned tasks, workflow, expected outputs, and supporting evidence.

Scope: The report documents the task specifications and a practical learning workflow. Exact experimental metrics, generated samples, and benchmark values are intentionally not fabricated because they were not included in the supplied task screenshots.

4.3 Technologies and Tools Used

Technology / Tool	Purpose
Python	Programming and experimentation
PyTorch / TensorFlow	Model implementation and training
Hugging Face Transformers	Working with GPT-2 and related language models
Diffusers / image-generation libraries	Working with pre-trained generative image models
NumPy / Pandas	Data preparation and manipulation
Matplotlib	Visualization and inspection
OpenCV / PIL	Image loading and preprocessing
Jupyter Notebook / VS Code	Development and documentation
Git / GitHub	Version control and project evidence

4.4 Methodology

- Read and understand the task objective and required inputs/outputs.
- Prepare the dataset or input images required for the selected task.
- Build or load the appropriate generative model.
- Run the generation or translation workflow and inspect the output.
- Evaluate the output qualitatively and record observations.
- Document the implementation, limitations, and evidence.

4.5 TASK 1 - TEXT GENERATION WITH GPT-2

Task objective: Train a model to generate coherent and contextually relevant text based on a given prompt. The supplied task specifies starting with GPT-2, a transformer language model, and fine-tuning it on a custom dataset so that generated text follows the style and structure represented in the training data.

Concept: GPT-2 is an autoregressive language model. During generation, the model repeatedly predicts the next token based on the sequence that has already been provided. Fine-tuning adapts a pre-trained model to the vocabulary, patterns, and writing characteristics of a selected dataset.

Inputs and Outputs

Item	Description
Input	Prompt text and a task-specific training dataset
Model	GPT-2 or a compatible transformer language model
Process	Tokenization, fine-tuning, prompting, generation
Output	Coherent text continuing the supplied prompt

4.5.1 GPT-2 Workflow

- Collect and clean the custom text dataset.
- Tokenize the dataset with the model tokenizer.
- Configure the GPT-2 model and training parameters.
- Fine-tune on the prepared dataset.
- Provide a prompt to the trained model.
- Generate text using controlled decoding settings.
- Inspect the generated text for coherence and relevance.

4.5.2 Key Considerations

Generation quality depends on the training data, tokenization, model capacity, training configuration, and decoding parameters. Sampling settings such as temperature and top-k/top-p can affect diversity and repetition. A practical evaluation should consider both fluency and whether the generated text follows the intended task style.

4.5.3 Implementation Notes

A typical implementation uses a tokenizer to convert text into token IDs, followed by supervised language-model fine-tuning. The training loop computes a next-token prediction loss and updates model weights. After training, prompts are passed through the model to generate continuations.

Expected Learning Outcomes

- Understand transformer-based language modelling at a practical level.
- Learn the relationship between prompts, tokenization, and generated continuations.
- Understand how fine-tuning changes the behaviour of a pre-trained language model.
- Recognize common generation issues such as repetition and off-topic continuation.

Evidence

The supplied task screenshot is included in the Project Screenshots section of this report.

4.6 TASK 2 - IMAGE GENERATION WITH PRE-TRAINED MODELS

Task objective: Use a pre-trained generative model to create images from text prompts. The supplied task mentions models such as DALL-E-mini or Stable Diffusion as examples of systems that can perform text-to-image generation.

Concept: Text-to-image models learn relationships between language descriptions and visual patterns. A prompt acts as a conditioning signal, and the model generates an image intended to match the described content, style, or composition.

Workflow

- Prepare a clear text prompt describing the desired image.
- Load a suitable pre-trained text-to-image model.
- Pass the prompt through the model pipeline.
- Generate one or more image samples.
- Inspect the output for prompt alignment and quality.
- Save the generated image and record the prompt used.

Expected Output

A generated image corresponding to the semantic content of the supplied text prompt. Exact examples or quality scores were not provided in the task material and therefore are not invented in this report.

4.7 TASK 3 - TEXT GENERATION WITH MARKOV CHAINS

Task objective: Implement a simple text-generation algorithm using Markov chains. The supplied task describes creating a statistical model that predicts the probability of a character or word based on one or more preceding elements.

Concept: A Markov model estimates the next state from a limited history of previous states. In text generation, the states may be characters or words. The transition probabilities are learned from an input corpus and then sampled to produce new sequences.

Algorithm / Steps

- Read and normalize the source text.
- Choose a Markov order, such as one previous word or several previous words.
- Count observed transitions from each context to the next token.
- Convert counts into probabilities or normalized frequencies.
- Choose an initial token or context.
- Sample subsequent tokens repeatedly to build a sequence.
- Stop after the requested length is reached.

4.7.1 Markov Chain Analysis

The main strength of a Markov text generator is simplicity. It can be implemented with dictionaries and frequency counts and can demonstrate sequence modelling without a neural network. Its limitation is that the model only remembers a restricted amount of context, so long-range meaning and global coherence are difficult to maintain.

Aspect	Observation
Model type	Statistical sequence model
State	Character or word
Conditioning	Previous one or more states
Training data	Source text corpus
Generation	Probabilistic next-state sampling
Main limitation	Limited context and long-range coherence

The task screenshot supplied by the user is included later as visual evidence.

4.8 TASK 4 - IMAGE-TO-IMAGE TRANSLATION WITH cGAN

Task objective: Implement an image-to-image translation model using a conditional generative adversarial network called pix2pix. The task focuses on learning a mapping from an input image to a corresponding target image.

Concept: In a cGAN, a generator creates candidate outputs conditioned on the input image, while a discriminator learns to distinguish generated pairs from real input-target pairs. Pix2pix is designed for paired image-to-image translation problems.

Workflow

- Prepare paired source and target images.
- Resize and normalize images to a consistent format.
- Train the conditional generator and discriminator together.
- Use adversarial and reconstruction-related objectives during training.
- Translate unseen input images with the trained generator.
- Compare the generated translation with the intended target domain.

4.8.1 cGAN / pix2pix Notes

The cGAN framework introduces a condition to the generation process, so the output is not created independently of the input. For pix2pix, the model learns from aligned image pairs, making the method suitable for tasks such as converting one visual representation into another when paired examples are available.

Key Learning Outcomes

- Understand the roles of the generator and discriminator.
- Understand why paired data is important for pix2pix.
- Observe the difference between unconditional generation and conditional translation.
- Recognize common challenges such as training instability and image-detail preservation.

4.9 TASK 5 - NEURAL STYLE TRANSFER

Task objective: Apply the artistic style of one image to the content of another image using neural style transfer. The supplied task gives the example of transferring the style of a famous painting onto a separate content image.

Concept: Neural style transfer separates two aspects of an image: content structure and stylistic patterns. A convolutional neural network can be used to extract feature representations, while optimization seeks an output image that preserves content information while matching selected style statistics.

Workflow

- Select a content image and a style image.
- Extract their feature representations with a CNN.
- Define content and style losses.
- Initialize an output image.
- Optimize the output to balance content and style objectives.
- Save and inspect the stylized result.

4.9.1 Neural Style Transfer Analysis

Component	Purpose
Content representation	Preserves the structure and main objects of the content image
Style representation	Captures textures, colors, and visual patterns of the style image
Content loss	Measures deviation from the content representation
Style loss	Measures deviation from the style representation
Optimization	Searches for an output image balancing both objectives

The visual result depends strongly on the relative weighting of content and style losses. Strong style weighting can make the output more painterly, while stronger content weighting can retain more of the original scene structure.

4.10 TESTING AND RESULTS

Testing for these tasks should verify both normal execution and the quality of generated output. Since the supplied material contains task descriptions rather than executed notebooks or numerical experiment results, this section records verification criteria rather than invented measurements.

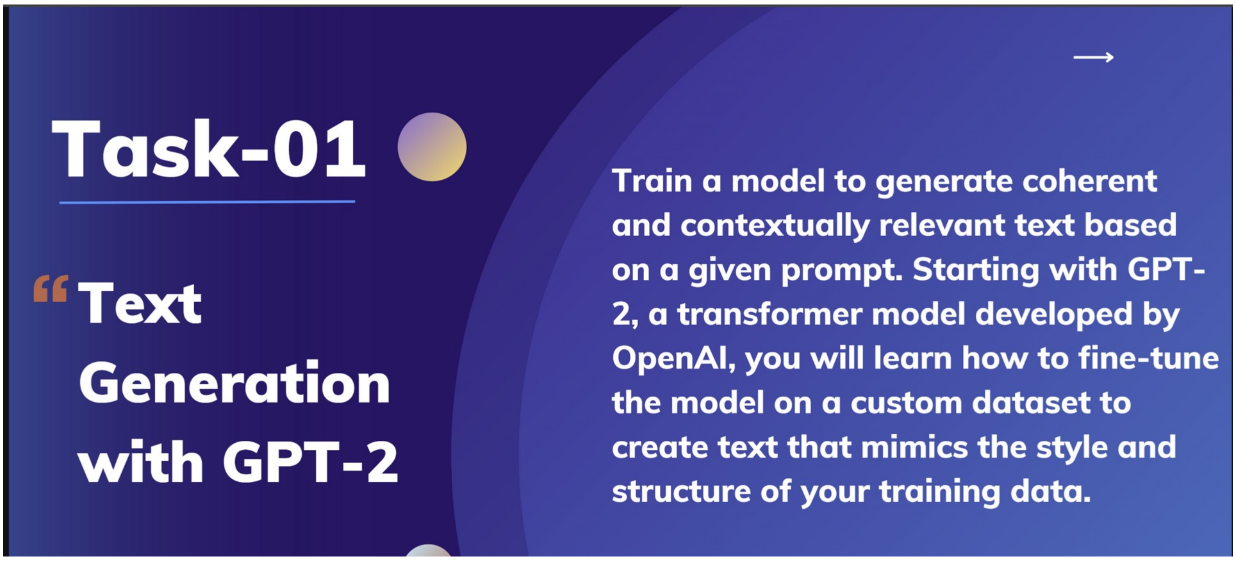
Task	Test	Expected Result	Status
GPT-2	Prompt generation	Contextually relevant continuation	To be verified after execution
Image Generation	Prompt-to-image request	Generated image matches prompt	To be verified after execution
Markov Chains	Sequence generation	Valid text sequence is produced	To be verified after execution
pix2pix cGAN	Image translation	Output is mapped toward target domain	To be verified after execution
Style Transfer	Content + style optimization	Stylized image is produced	To be verified after execution

4.11 LEARNING OUTCOMES

- Understanding of several Generative AI paradigms.
- Practical exposure to text and image generation workflows.
- Understanding of probabilistic sequence generation.
- Awareness of conditional adversarial learning for paired image translation.
- Understanding of content/style separation in visual generation.
- Improved ability to document model workflows and limitations.

4.13 PROJECT SCREENSHOTS

Task 1 - Text Generation with GPT-2



Task evidence supplied by the student.

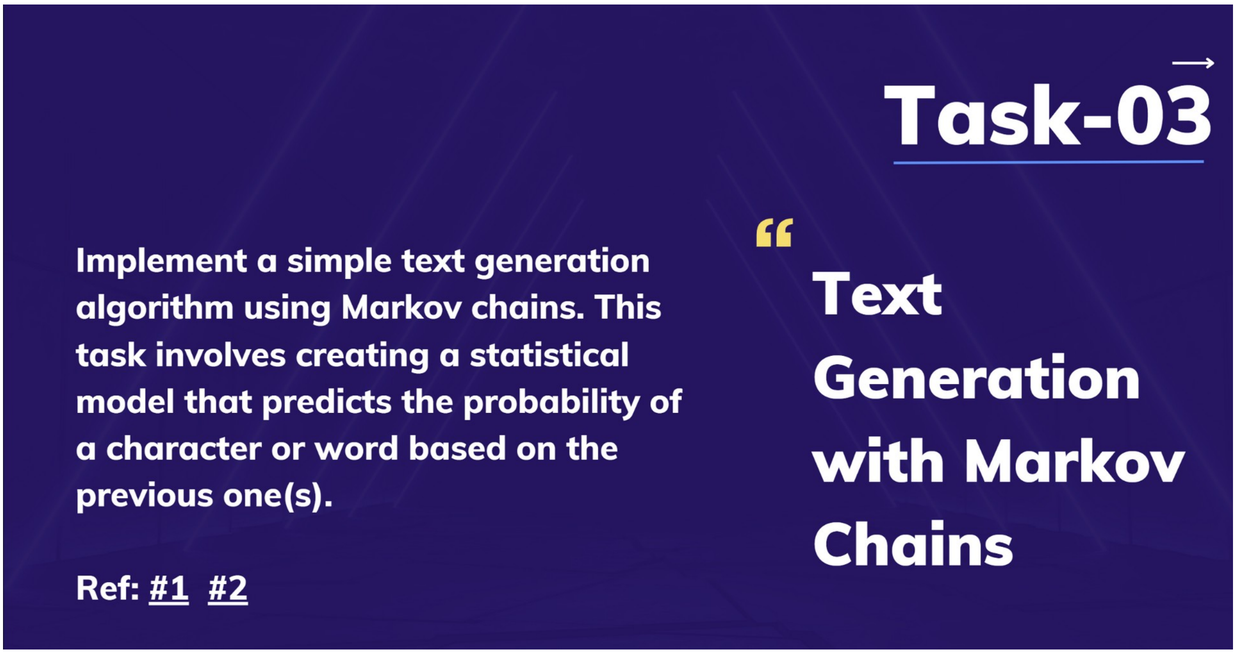
4.13 PROJECT SCREENSHOTS

Task 2 - Image Generation with Pre-trained Models

Task evidence supplied by the student.

4.13 PROJECT SCREENSHOTS

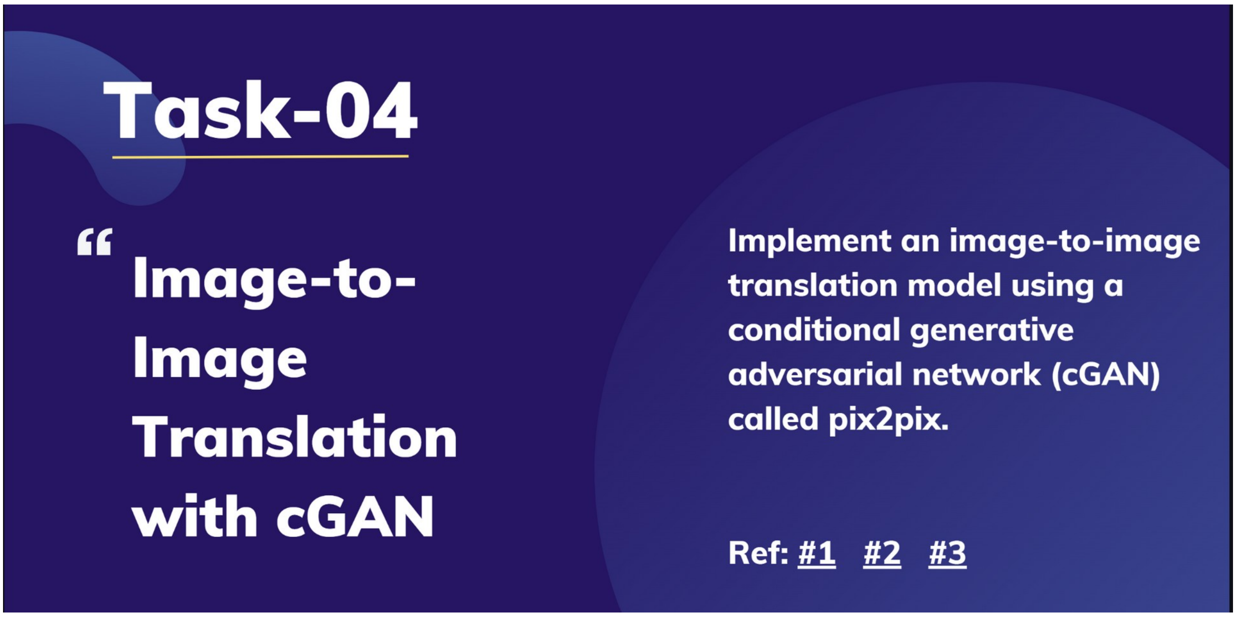
Task 3 - Text Generation with Markov Chains



Task evidence supplied by the student.

4.13 PROJECT SCREENSHOTS

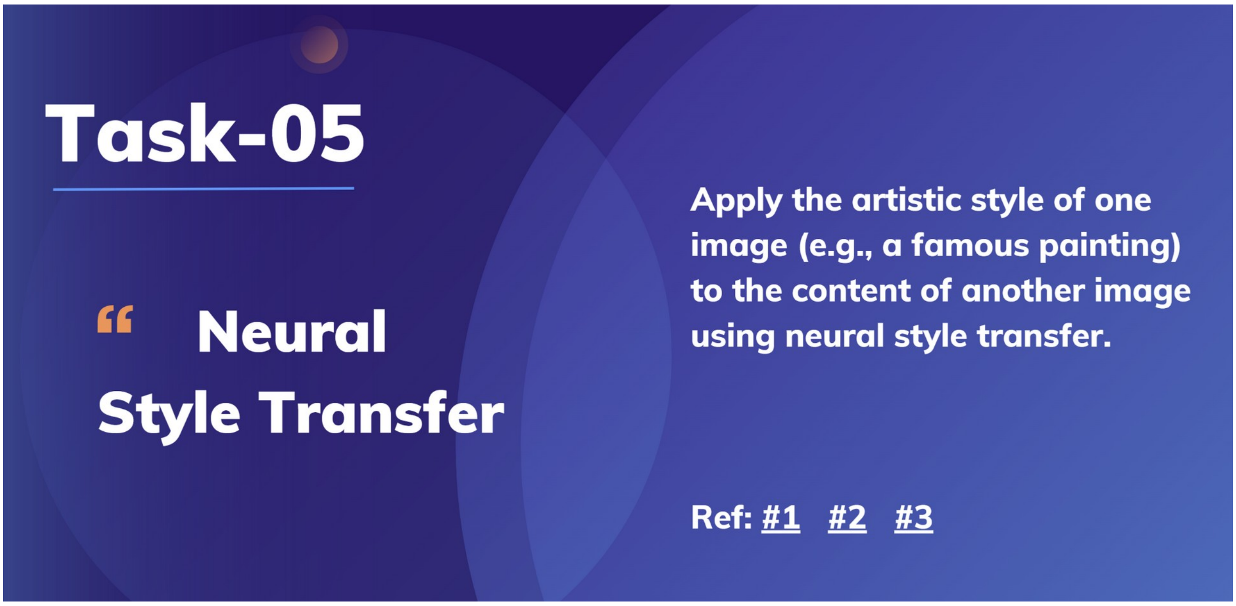
Task 4 - Image-to-Image Translation with cGAN



Task evidence supplied by the student.

4.13 PROJECT SCREENSHOTS

Task 5 - Neural Style Transfer



Task evidence supplied by the student.

5. BIBLIOGRAPHY / REFERENCES

1. Radford et al., GPT-2 / transformer language-model literature and documentation.

2. Hugging Face Transformers documentation for causal language modelling.

3. Hugging Face Diffusers documentation for text-to-image generation pipelines.

4. Goodfellow et al., Generative Adversarial Networks and related cGAN literature.

5. Isola et al., Pix2pix: Image-to-Image Translation with Conditional Adversarial Networks.

6. Neural style transfer research and deep feature visualization literature.

7. Python documentation and standard scientific-computing libraries.

8. The five task specifications and visual references supplied with the internship assignment.

Report Information

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Domain	Generative AI